

The main characteristic of a swarm is that the robots are spread out in an environment and each swarm robot 'talks' to another robot within the same group. Soccer robots are another set of robots that communicate in a similar fashion amongst respective team robots and succeed in scoring a goal."

Swarm robots usually communicate using wireless technologies like infrared and radio frequency (RF). One such option is RF PRO, made by Robosoft Systems. RF PRO is a bidirectional radio frequency module used by robots to exchange data with each other. It works on 2.4 GHz and

stant in the decision-making process. Any delay in its implementation may be even more disastrous than the crisis itself. Intelligence is of no use if it is slow," says Dr Venkatesh.

"Present efforts towards an intelligent robotic system aim to localise the thought process in such a way that it avoids extensive processing time. One simple solution is the sense-action methodology rather than sense-process-action. The best way is to implement parallel processing techniques that can simulate nature's reflex action technology. It is the reflex action that saves our hand from getting roasted

present, perform better in controlled environment.

While industrial robots remain the most time-tested clan of our machine friends, the demand even for these is limited in India. Cost is just one of the factors. The industrial economics of our country is also such. With resistance from the existing employees to go for such automation systems due to fear of losing jobs on one hand and the availability of cheap manpower on the other hand, neither the workers nor the companies have any strong motivation to opt for robots. Only certain forward-looking companies or industries involving very high-precision or hazardous work go in for robots.

Dr Venkatesh notes that in India, R&D on the industrial robotics front is confined mainly to customisation of solutions. However, Azad notes that all major international brands in automation and robotics—including Motoman, Fanuc, Panasonic, Siemens, ABB and Kuka—have a good presence in India. He feels that it is the domestic or service bots like Romba and simpler systems for small-scale industries that are missing.

Apart from industrial use, robots are gaining ground in medicine, agriculture and education as well. We are all aware of the Da Vinci system that made remote surgeries using robots possible. A couple of years ago, Energid developed a system for the American army to perform highly intensified ultrasound (HIFU) tele-surgery in order to address warfront casualties in a master-slave mode with force feedback. It also developed for the US Department of Agriculture a patented, tongue-like robot for harvesting citrus fruits. The company's Indian division has completely designed and developed Cyton—a humanoid robot arm that is capable of mimicking all the motions of the human arm. It can play cricket, cut a cake, handle bottles, clean your table and even make a cup of tea for you. Jayakrishnan reveals that the Canadian Space Agency is going to fit this manipulator in its spacecraft.

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has a range of 50 metres. Up to 65,000 modules can exchange data with each other without any interference between them.

**Artificial intelligence.** Advancements in the space of artificial intelligence (AI) continue to complement the improved sensory powers of robots. Azad says that as far as AI and robotics go, the focus is more on learning algorithms. There is a lot of work on self-modelling and adapting to changes in the environment and robot structure. For example, the failure of one limb of a robot will require it to re-calculate a complete new walking algorithm for it to be able to walk again.

Dr Venkatesh points out an inherent problem that is yet to be tackled. AI involves intensive computing and computational complexities, resulting in a time delay in the sense-process-action sequence of robots. It may be superfluous to expect human intelligence simulated and implemented in a robot that is quite slow in its movement, thought and action.

"How do you visualise a robot that has a time delay built into it due to machine inertia? Time is very impor-

tant in a fire when we accidentally touch it," he adds.

**Open Source software and hardware designs.** Very reliable Open Source real-time operating systems and single-board computer designs are available today. These are updated frequently, and include all the latest capabilities. Plus, they are highly-customisable—a critical benefit for robotics engineers.

**Other technologies.** Current generation of micro-actuators plays an important role in advanced robots used for invasive medical applications, flight manipulation, etc. Nanopositioning actuators are used in probe microscopes. MEMS actuators are indispensable for implementation of various robotic sensors. Actuators mimicking soft artificial muscles are used in robots that mingle with humans, including those used in healthcare.

## Where do we see these robots?

World-over industrial robots are more common than domestic ones. This is perhaps because robots, at